

Impact of a grid-connected PV system application in a bioclimatic house toward the zero energy status in the north of Algeria



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ABSTRACT

Nowadays, it is well recognized at the international level that the building sector is one of the biggest energy consumers, which made the intervention in this sector the greatest potential for energy consumption reduction and greenhouse emissions mitigation. In Algeria, building sector is responsible of more than 40% of the final energy consumption. Solar energy systems integrated in buildings can be an important solution to this dual problem.

The aim of this paper is to estimate the energy performance of a typical single-family bioclimatic house equipped with a solar heating system located in the north Algeria (Algiers) and to investigate its conversion into a zero energy house by installing a grid connected PV system.

To achieve this goal, a numerical model of the bioclimatic house coupled with the solar heating system is developed and validated with experimental data. Then, the validated model is used to optimize the adequate size of the solar heating system and to determine its contribution in the production of thermal energy for both DHW preparation and space heating of the house. Finally, a PV system is sized to supply, in parallel with the utility grid, the house with electricity for lighting, appliance and auxiliary electric heater. Furthermore, an economic analysis is made to show the profitability of integration of grid connected PV and solar thermal systems in building.

Results show that the conversion of a bioclimatic house into a ZEH using a grid connected PV system, in the north of Algeria weather conditions, present high level of feasibility. However, the return on investment is very high due mainly to the high cost of solar energy components and the very low costs of conventional energy in Algeria.

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1. Introduction

The Building sector is one of the major consumers of energy in the world. According to a recent survey, the energy consumed in buildings accounts for more than 42% of the final energy consumption in Algeria due mainly to population growth and urbanism [1]. To reduce energy consumption of this sector, Algeria developed a thermal regulation for residential buildings including three Regulatory Technical Document (DTR), which allow a saving of about 20–30% on energy consumption for space heating [2–5].

The bioclimatic design is increasing in popularity as an intelligent way to build energy efficient buildings, as well as to increase indoor comfort inside the house over the entire year. Incorporating energy efficiency measures into the design step of new buildings

and undertaking retrofits in existing buildings have been investigated in many recent papers [6–10].

Moreover, it has been demonstrated that the remaining energy demand of these energy efficient buildings, can completely been met by renewable sources [11]. As a result, several high energy performance building (HEPB) concepts have been proposed, from low-energy building through passive building and zero-energy building to positive energy building and even autonomous building [12].

Grid connected zero energy building (ZEB) or zero energy home (ZEH), called Net ZEB or Net ZEH, is the most studied approach in literature. This grid connected ZEH uses the utility power grid for storage-delivering energy to the grid when the PV system produces more energy than the home uses and draws from the grid when the PV system produces less energy than the house needs.

To achieve Net ZEH status, different energy solutions have been studied. In all cases, the first phase is to improve the energy efficiency of building by passive strategies in order to reduce heating, cooling and electrical energy loads. The second phase consists in

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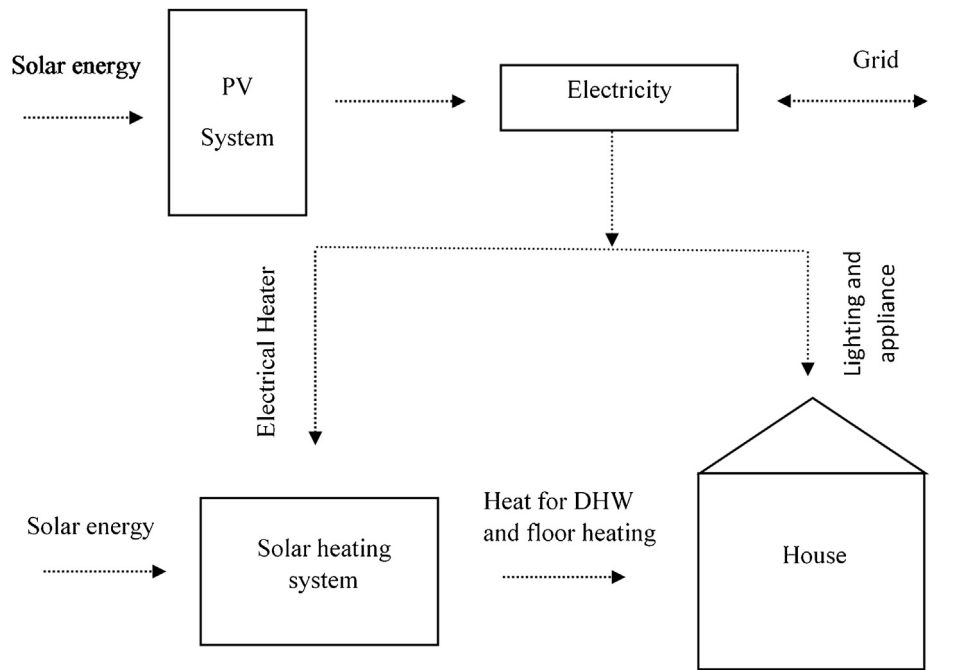


Fig. 1. The house, its heating system, and PV system under investigation.

adding renewable energy systems to meet the remaining energy demand. These systems include:

- A solar PV system, water-to-water heat pump and a ground heat exchanger for space heating system, lighting and appliance, and to heat DHW [13].
- Small wind turbine and PV array to generate electricity for lighting, appliance, auxiliary heating for SDHW and energy consumption in floor heating system [14].
- Reducing the heating and cooling energy loads by passive means to the point of eliminating the need for an auxiliary heating or cooling system, an operationally zero energy house can be created by installing a photovoltaic system [15].
- A solar combisystem that uses solar energy, captured by solar collectors, to supply heat for both radiant floor heating system

and DHW tank. On the other hand, a PV system for electric power generation is used. [16].

In this paper, the last solution is adopted to investigate the impact of a grid connected PV system utilization in a typical single-family bioclimatic house located in the north of Algeria towards its conversion into zero energy house (ZEH).

2. Methodology

The principle of ZEH is shown in Fig. 1. As shown, the solar energy systems include a solar heating system for both DHW preparation and space heating. The electrical energy needed for domestic electrical consumption (lighting and appliance) and auxiliary electrical heater is provided from a grid-connected PV system.

This work is structured as follows:

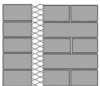
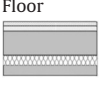
- First, a numerical model of the solar heating system, coupled with the bioclimatic house, was developed under TRNSYS software, and validated with experimental data.
- Then, based on the validated model, a parametric study was conducted to optimize the size of the solar heating system components and establish the energy balance of the house with and without optimized solar heating system.
- Finally, a grid-connected PV system is sized to supply, in parallel with the utility grid, the house with electricity. In the simulation, the electricity required for the auxiliary electric heater was included to achieve the goal of zero energy house status. Furthermore, an economic analysis is made to show the main barriers for the application of the grid connected PV system.

TRNSYS software [17] was used to estimate the heat demand for space heating and DHW preparation as well as space cooling load. and HOMER, [18] was used in order to calculate the electricity produced from the photovoltaic system under investigation.



Fig. 2. The prototype bioclimatic house.

Table 1
U values of the building envelope.

Layers	Composition	Thickness	U value (W/m ² K)
	Stabilized earth blocs	0.14	0.36
	Polystyrene	0.09	
	Stabilized earth brick	0.29	
	Concrete	0.05	0.23
	Polystyrene	0.06	
	Concrete	0.15	
	Mortar	0.03	
	Tiling	0.02	
	Mortar	0.03	0.54
	Polystyrene	0.16	
	Concrete	0.8	
	Plaster	0.3	
	Double glazing		2.22

3. Experimental study

3.1. Bioclimatic house characteristics

A detached bioclimatic house with a 90-m² floor area, realized in the framework of MED-ENEC Project, was selected in order to analyze the feasibility of zero energy house in Mediterranean climate conditions (Fig. 2). The house walls are built with stabilized earth blocs and the external walls and floor are well insulated, in addition to PVC double glazed windows and shading devices. The house is equipped with a solar heating system for Domestic Hot Water (DHW) and space heating. Thermal properties of walls are listed in Table 1.

3.2. Solar heating system

Solar combisystem is a solar heating system which can be used for both space heating and hot water production. One of the most comprehensive studies on solar combisystem was performed within the frame of IEA-SHC-Task 13 and 26 (IEA 2007) by comparing 21 different configurations [19]. The configuration and the main components of the solar heating system studied in this work are shown in Fig. 3: solar collectors (1) installed on the top of the pitched roof (45°) and in accordance with the local latitude of the site, hot water tank (2), under heating floor (7).

The operation of the solar heating system is as follows: The three-way valve (5) is under the control of a differential controller (6) in the pumping station. Priority is given to the load (space heating). Thus, the system operates as a Direct Solar Floor. The collector array is directly connected to the heating floor; the concrete slabs combine the functions of space heating storage and heat exchanger. The water heated in the solar collector, flows through the heating floor and then flows back to the solar collector. If the temperature of the unfavorable zone of the house, which is controlled by a thermostat (3) located on the ground floor, is higher than the set-point temperature, the single speed pump of the system starts to operate and the hot water is oriented toward the storage tank for DHW

Table 2
Mean relative error.

	T _{min} (°C)	T _{mean}	T _{max} (°C)	MRE (%)
Collector	00	0.08	0.46	1%
Room1	0.03	0.44	1.55	2.30%
Room2	0.01	0.20	0.82	1.61%
Living room	0.05	0.53	1.48	3.14%

preparation. The cold water from the city main enters at the bottom of the storage tank. A tempering valve (8) is used to enable the control of DHW temperature at around 45 °C at the user-end. An auxiliary water heater is used to ensure a correct water temperature for DHW.

3.3. Data acquisition

A data-acquisition system was set up to register the experimental variables, as shown in Fig. 4. This system consists of a computer and a Keithley digital multi-meter2700. A set of type-K thermocouples is used to measure temperatures at several locations. All thermocouples were calibrated against a laboratory thermometer in a water bath at various temperatures. Global Solar radiation is measured using KIPP&ZONEN pyranometer mounted in accordance with the slope of the collector (45°). The indoor air temperature in different zones of the house is measured using thermo-hygrometers (accuracy ± 0.5 °C).

The solar radiation and the outdoor temperature are recorded at 30 min intervals during the experimental period.

3.4. Experimental results

In order to study the effect of passive and active measures, different scenarios are studied. First, We consider that the floor heating system is activated only in the Room1 (active solar gain), the Living-room benefited from only the passive solar through a large window facing south and in the Room2, there is any solar gain.

Fig. 5 reports experiment results for four days in December (21th–24th of 2012). December is one of the coldest months of the year in Algeria. All the four days are sunny. The maximum solar radiation during the monitoring period vary between 805 W/m² in the first day and 915 W/m² in the last day. It can be noticed clearly the impact of solar heating system on the indoor air temperature of room 1. The indoor temperature gradually increases, passing from 15 °C in the first day to 20 °C in the fourth day when the outdoor temperature is 17 °C.

The results reveals also the contribution of passive solar gain in the improvement of indoor comfort during cold period. Due to the large glass area, indoor temperature in this zone reach 17.5 °C in daytime and stay upper than 16 °C in nighttime. This is explained by the fact that insulation and thermal inertia of walls significantly reduces temperature fluctuations of indoor air. For the room 2, as the solar gain is absent and this zone is located in the north, the indoor temperature decreases from 15 °C to 13.5 °C.

3.5. Validation of the model

The recorded weather data were used as inputs of the TRNSYS solar system model. To test the accuracy of the model, the mean relative error (MRE) model is used [20]:

$$MRE = \left| \frac{\sum_{i=1}^n (T_{simulated} - T_{measured})}{n T_{measured}} \right| \times 100\% \quad (1)$$

The measured and simulated hourly temperatures at the output of the thermal collector is shown in Fig. 6. Table 2 shows that the difference of the MRE between the simulated and the measured val-

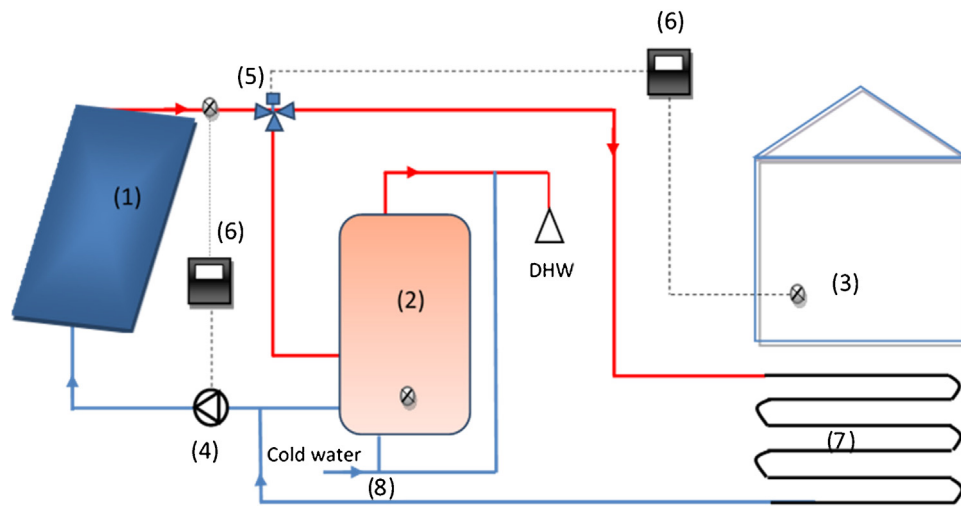


Fig. 3. Scheme of the solar heating system.

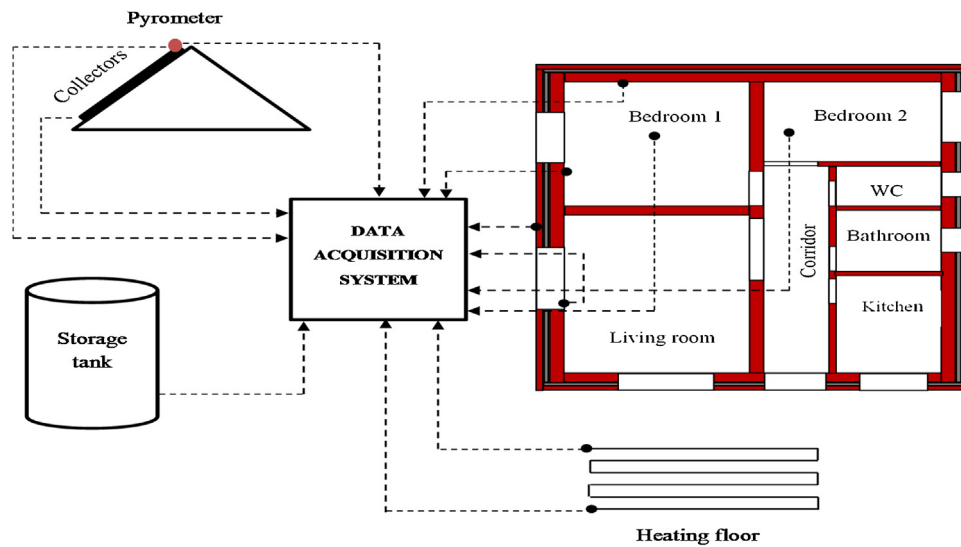


Fig. 4. Scheme of data acquisition system.

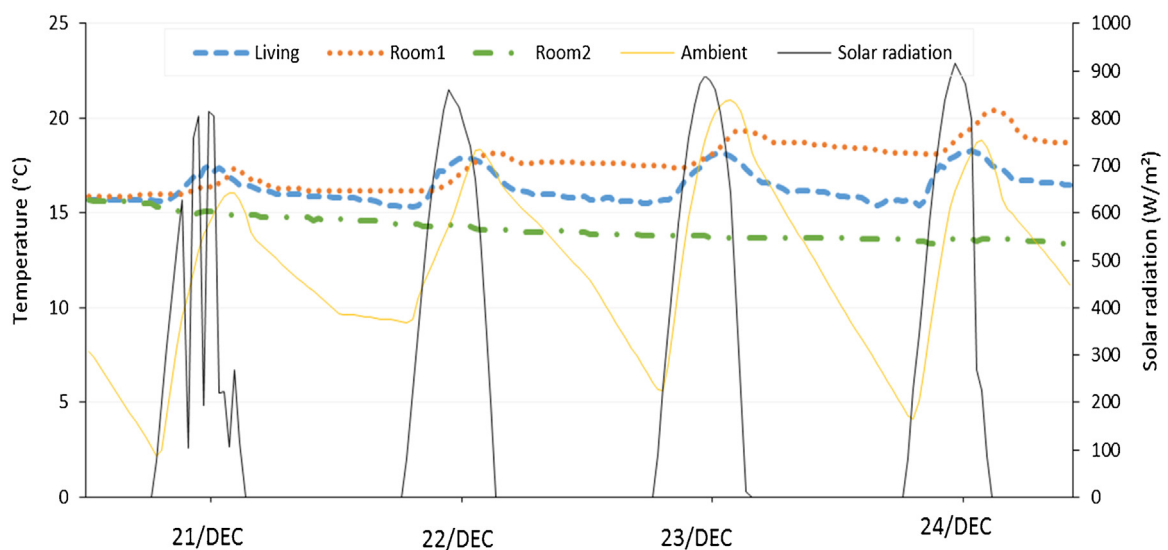


Fig. 5. Measured temperature of interior air of the house.

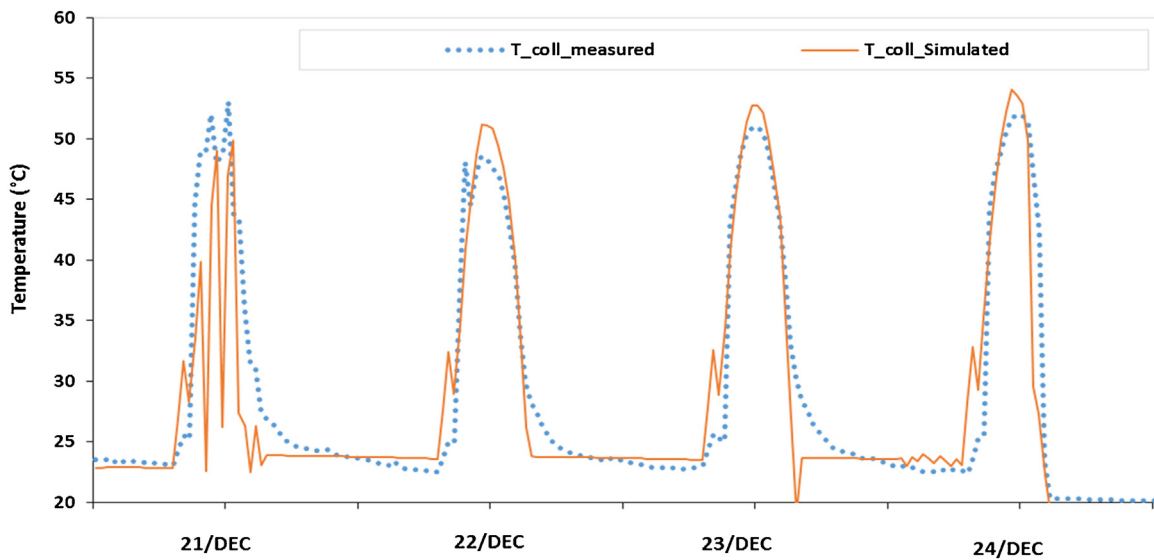


Fig. 6. Measured and simulated temperatures of the outlet collector fluid.

ues are less than 1%, the minimum absolute temperature difference 0 °C and the maximum absolute temperature difference 0.46 °C.

The hourly indoor temperatures (simulated and measured) in the Room1, Room2 and Living-room are shown in Figs. 7 and 8. We can noticed that there is a similar change trend between the simulated and the measured values. The MRE is 2.30%, 1.61% and 3.14% in the Room1, Room2 and Living-room, respectively. The minimum and the maximum absolute temperatures differences are 0.03 °C and 1.55 °C, in the Room1, 0.03 °C and 0.82 °C in the Room2 and 0.05 °C and 1.48 °C in Living-room, respectively. In general, the simulated results agreed well with the experiment results.

4. Energy performance analysis of the bioclimatic house

In order to estimate the annual energy balance, it is assumed that the house is occupied by four people from 5:00 pm to 8:00am, and one person from 8:00 am to 5:00 p.m. The gains from occupancy are 115 W/pers (75 W sensible and 40 W latent).

Fridays and Saturdays, four persons presence is assumed throughout the day. The temperature set points for the zone were set to 20–26 °C from 7:00 am to 10:00 pm and during sleeping hours a wider thermal comfort range of 18–26 °C was implemented as a lower temperature can be tolerated in bedrooms.

The current total average Algerian individual domestic hot water usage is taken to be 50 l/day [21]. Therefore, the total domestic hot water consumption is 200 l/day. Hourly DHW profile is illustrated in Fig. 9. The mains water is assumed to remain constant at 15 °C and the set point for DHW is 45 °C.

4.1. Energy balance without solar heating system

4.1.1. Space heating and DHW demand

The monthly energy demand of the house without solar heating system contribution is shown in Fig. 10. There are two loads to be covered by the solar heating system. The thermal energy for DHW preparation and space heating requirement. The heat demand for space heating is absent from Mai to October and important during

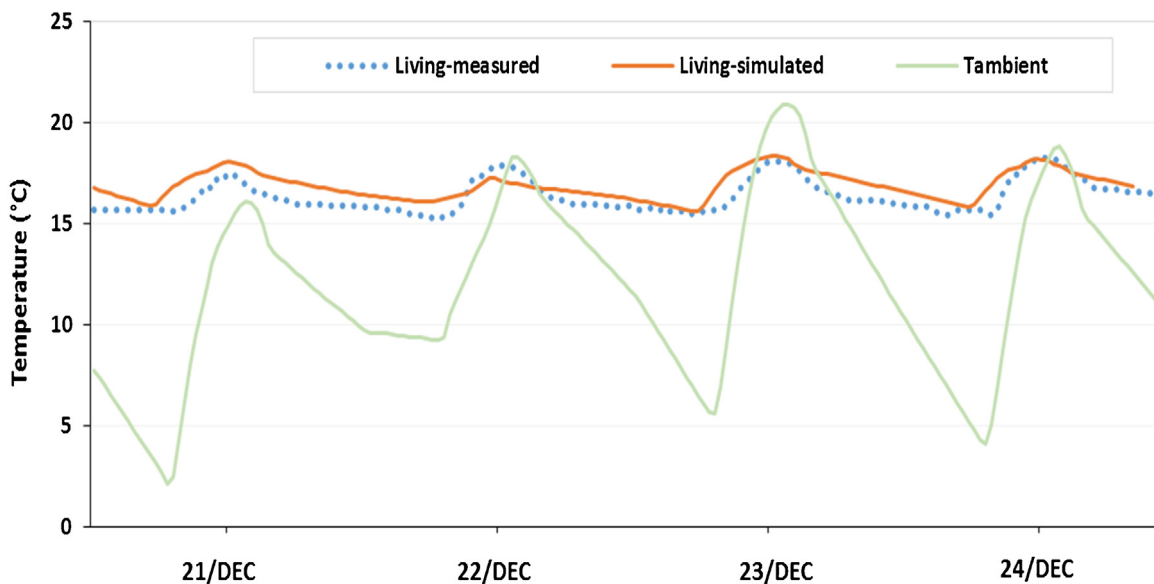


Fig. 7. Measured and simulated temperatures of the interior air of the living room.

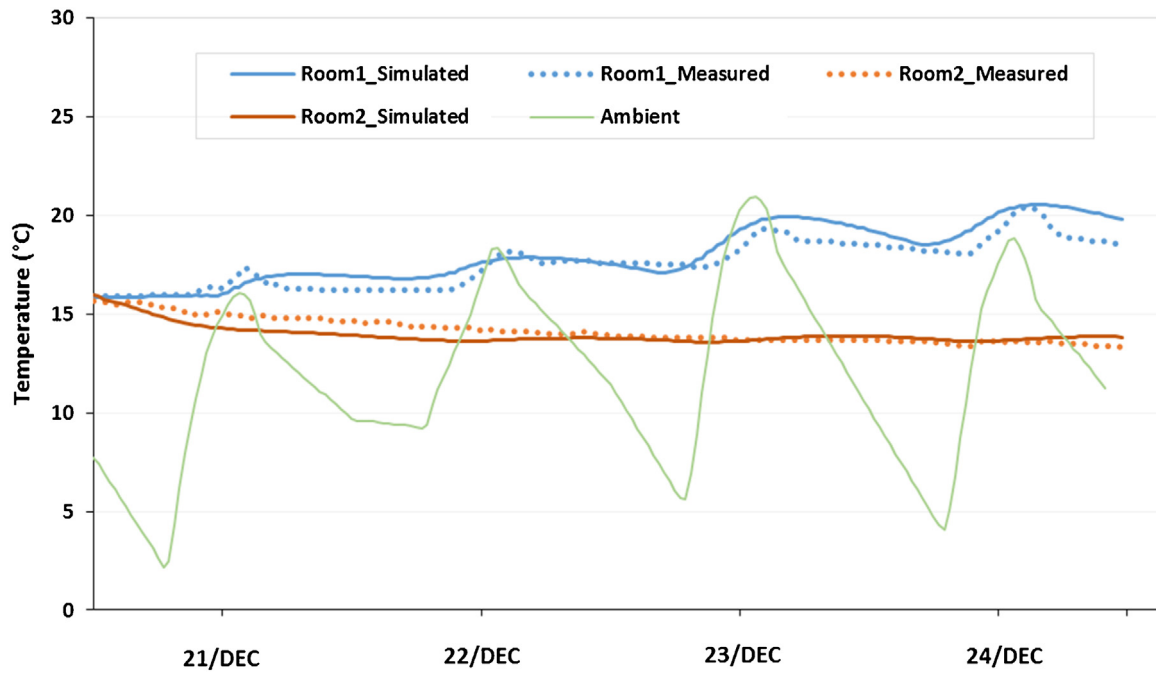


Fig. 8. Measured and simulated temperatures of the interior air of Room1 and Room2.

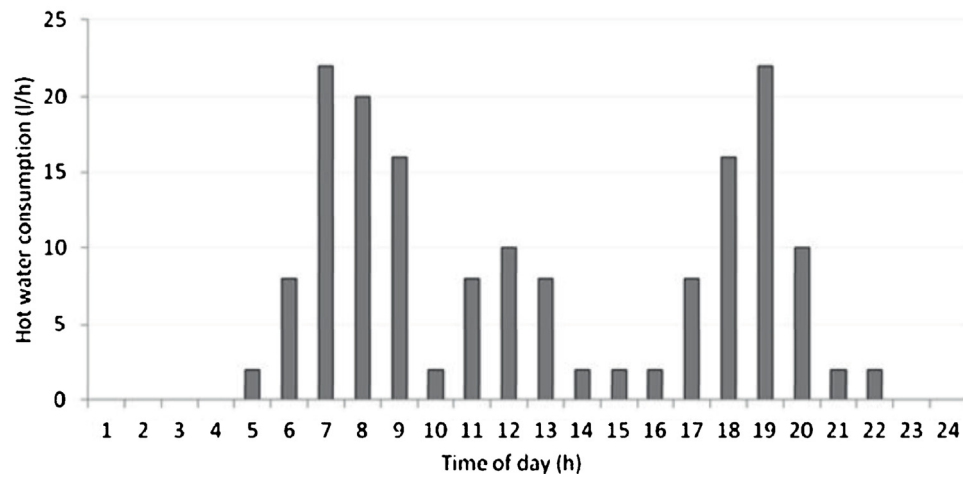


Fig. 9. Daily hot water consumption profile (200l/day).

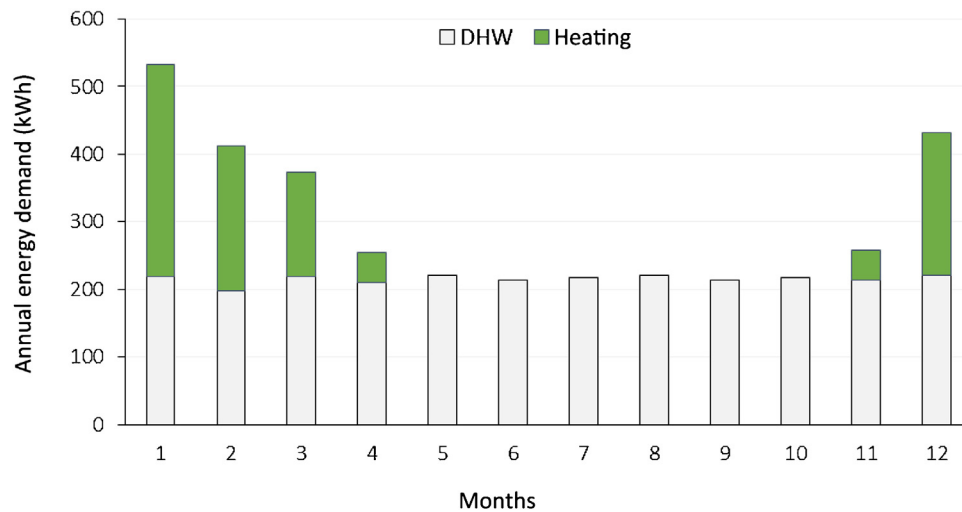


Fig. 10. Monthly energy demand.

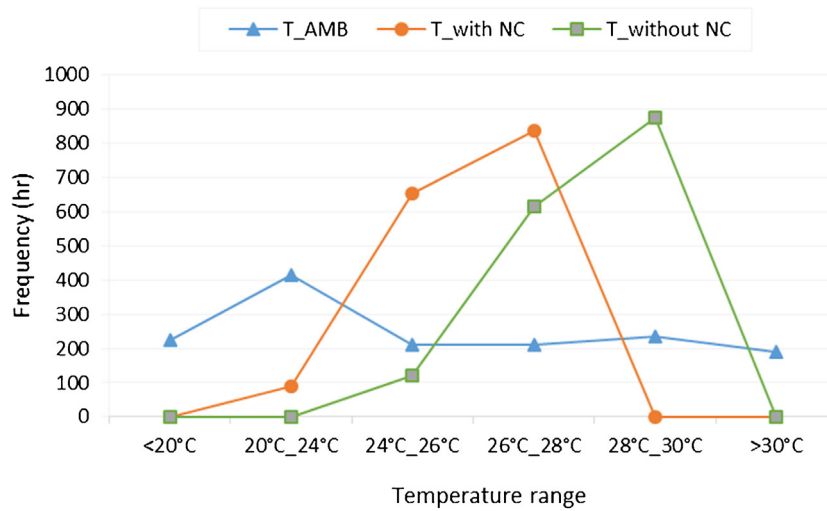


Fig. 11. Thermal comfort in July and August.

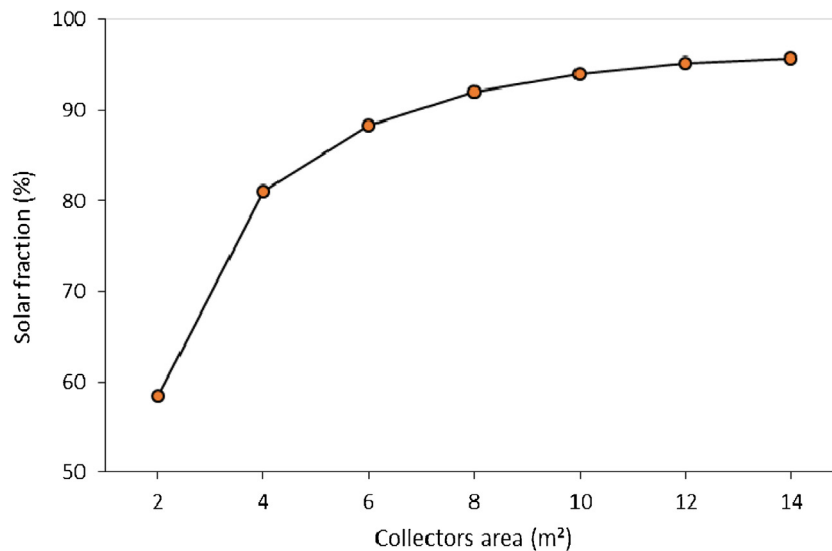


Fig. 12. Variation of the annual solar fraction for different collector areas.

cold season. The space heating demand increase from December to January and decrease from January to April. However, the heat demand for DHW is very similar for all the months of the year (about 228 kWh/month).

The annual heat required for the preparation of the DHW represent the great fraction (72%, 2581.53 kWh), due to the fact that the DHW is used over all the year. Thanks to the bioclimatic designs that decrease the heating energy demand of the house, which represents only 28% (982.24 kWh) of the total heat demand.

4.1.2. Summer thermal comfort

Thermal comfort has been evaluated, basing on the computed hourly indoor temperatures during July and August, the hottest months in year. We considered that the house is naturally ventilated during nighttime by opening the windows from 24:00 pm to 5:00 a.m. The infiltration rate is 0.6 volume per hour during night cooling and 0.2 volume per hour during daytime. We consider also that the shading device blocks 80% of incoming solar radiation for southern windows.

A temperature frequency plot of the resulting indoor temperatures of the house with and without night cooling is presented in Fig. 11. The thermal comfort range is between 20°C and 28°C.

Table 3

Indoor thermal comfort in the house with and without night cooling.

	20°C_24°C	24°C_26°C	26°C_28°C	28°C_30°C	>30°C
T. without NC	0	120	616	874	0
T. with NC	138	710	730	0	0
T_AMB	415	211	210	235	190

There is a clearly difference between these two cases. The number of temperatures out the thermal comfort range in the case without night cooling illustrates the higher energy need to maintain the house within the thermal comfort range. In this case, the number of hours outside the thermal comfort range is significantly higher. As shown in Table 3, the indoor temperatures are above 26°C for 616 h and above 28°C for 874 h of the year. In the case with night cooling, overheating period is very limited essentially thanks to night ventilation. Indeed, the indoor temperature for this house does not exceed 28°C in summer, which is in accordance with the results of the experimental study conducted by [22]. This study demonstrated that the use of shading device and nocturnal ventilation are sufficient to provide thermal comfort in summer.

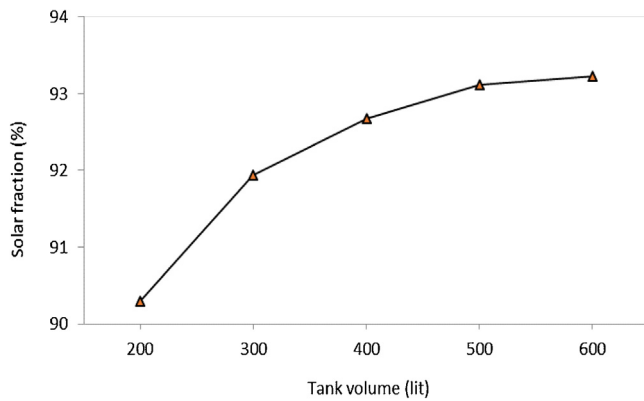


Fig. 13. Variation of the annual solar fraction for different tank volume.

4.2. Energy balance with solar heating system

4.2.1. Optimization of solar heating system parameters

The solar fraction is an optimization parameter that manifests the overall performance of the entire system. Thus, compared to other parameters, which indicate the performance of a single component, the solar fraction is the better indicator of the system performance [23]. The solar fraction is the fraction of the total hot water energy that is supplied by solar system, is calculated using the equation [24].

$$SF = \frac{(Q_{load} - Q_{aux})}{Q_{load}} \quad (2)$$

4.2.1.1. Effect of collector area. In order to determine the adequate collector area, five collector areas that 2, 4, 6, 8, 10, 12 and 14 m² are considered. The initial value of the hot water tank volume-to-collector area ratio is set equal to 75 l/m², which is the base value for the f-chart method [25]. The recommended range of the collector mass flow rate-to-collector area ratio in the literature [26,27] is initially set equal to 40 kg h⁻¹ m². The annual solar fraction changes for each collector area is presented in Fig. 12. The results show that increasing the collector area from 2 to 10 m² increase sharply the annual solar fraction of the system. An increasing of 1.5% in solar fraction is recorded between 10 and 14 m². Considering this little increasing in solar fraction after 10 m² and the cost of solar thermal collector, the adequate collector area is 10 m² for the present study. The solar fraction at this size reach 93.99%.

4.2.1.2. Effect of tank volume. The effect of the tank volume on the solar fraction of the thermal system is studied for various tank volume (from 200–600 l). The impact of the tank volume on the annual solar fraction of the solar system is shown in Fig. 13. As shown, the solar fraction increases rapidly, as the tank volume increases from 200 to 300 l. This means that the thermal energy stored at the storage tank is higher than the heat losses. For the tank volume values between 300 and 500 l, the increase in solar fraction is gradual. Beyond 500 l, the increase in the annual solar fraction is insignificant, which is likely due to an increase in heat losses from the storage tank, since the tank become larger. Considering the optimal collector area of 10 m², the optimal value of 500 l represents the optimal capacity of the storage tank that permit to reach of the maximum rate of solar fraction of 93.12%.

4.2.1.3. Effect of collector mass flow rate. The effect of collector's mass flow rate on the annual solar fraction was simulated relatively to the optimal collector area of 10 m² and tank volume of 500 l. Different mass flow rate ranging from 50 to 250 kg h⁻¹ are considered, as shown in Fig. 14. The results show that the solar fraction rises

Table 4

Technical specification of solar PV module (CEM200M-72).

Parameter	Specifications
Nominal maximum power	200 W _p
Optimum operating voltage	36.5 V
Optimum operating current	5.48 A
Open circuit voltage	44.5 V
Short circuit current	5.92 A
Operating temperature	−40 °C ~ +85 °C
Normal operating cell temperature	45 ± 2 °C
Dimension	1580 mm × 808 mm × 45 mm

rapidly as the mass flow rate increases from 40 to 150 kg h⁻¹. It increase from 91.39% at kg h⁻¹ to 93.12% at 150 kg h⁻¹. The rate of heat transfer from the thermal collector to the working fluid is directly proportional to mass flow rate. This mean that the flow rate removes efficiently the heat from the thermal collector. From this value, the solar fraction starts to decrease with further increase in mass flow rate because of the energy removed from the collector and delivered to the tank decreases. Hence, to promise a maximum value of solar fraction, the flow rate traversing 10 m² of thermal collector should be equal to the optimal value of about 150 kg h⁻¹.

4.2.2. Contribution of the solar heating system

The monthly heat supplied by the electrical heater for DHW preparation and auxiliary heat needed for space heating and solar fraction are shown in Fig. 15. we can noticed that the solar heating system is able to provide the most part of heat energy requirements in summer, since the space heating loads is absent. A solar fraction of 97% and 96% is reached in July and August, respectively. During offseason, an important energy demand is covered by the heating system. For example, a solar fraction of 77.03% and 68.32% can be achieved in April and November, respectively. During cold months, as the solar collectors could be used to provide heat demand for space heating and DHW, the heat output from the heating system is insufficient to cover these loads. The most supplied heat was oriented toward the space heating. During this period, the use of classical heater is required. The minimum solar fraction is 44% in January. We can noticed also that as the month is cold, as the heat supplied by the classical heater is high.

The annual heat produced by the solar heating system and classical heater shows that the most annual heat demand can be covered by solar heat gain output from the heating system, which is represent 93% (3362 kWh) of the total energy supplied. The annual heat supplied by the classical heater represents only 7% (191 kWh).

5. Contribution of the PV system

The PV system would produce electrical energy to satisfy needs for electrical energy of the house. The house uses the utility power grid for storage-delivering energy to the grid when the PV system produces more energy than the home uses and draws from the grid when the PV system produces less energy than the house needs. The scheme of a grid-connected PV system chosen in this study is shown in Fig. 16. It is comprised of a PV array, an inverter, a meter to measure the energy exported to the grid and the energy imported from the grid. The solar PV module is 72-cell (6 × 12) mono-crystalline with a rated power of 200 W_p and manufactured by the Algerian group, Condor. The detailed technical specifications of this model are presented in Table 4 and of the inverter model used to supply the required AC power output are presented in Table 5. The array slope is set to 45° in accordance with local latitude (36.13 N) and can be installed on the flat roof.

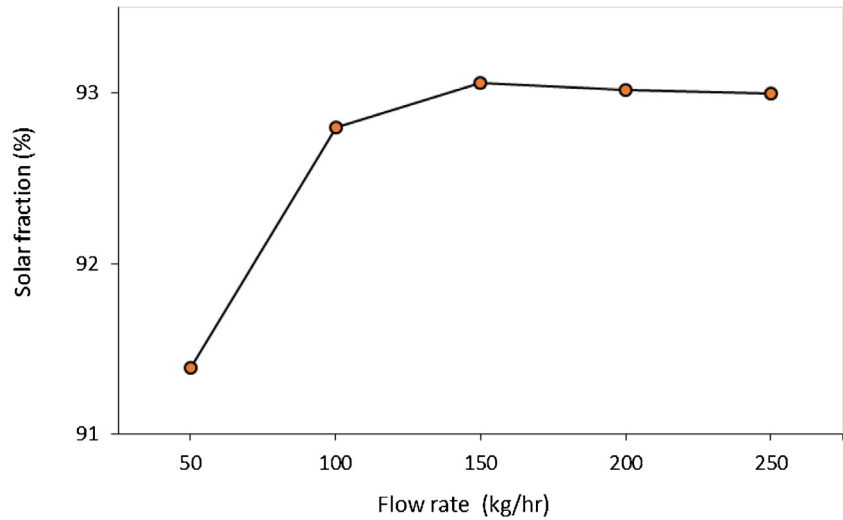


Fig. 14. Variation of the annual solar fraction for different collector's mass flow rate.

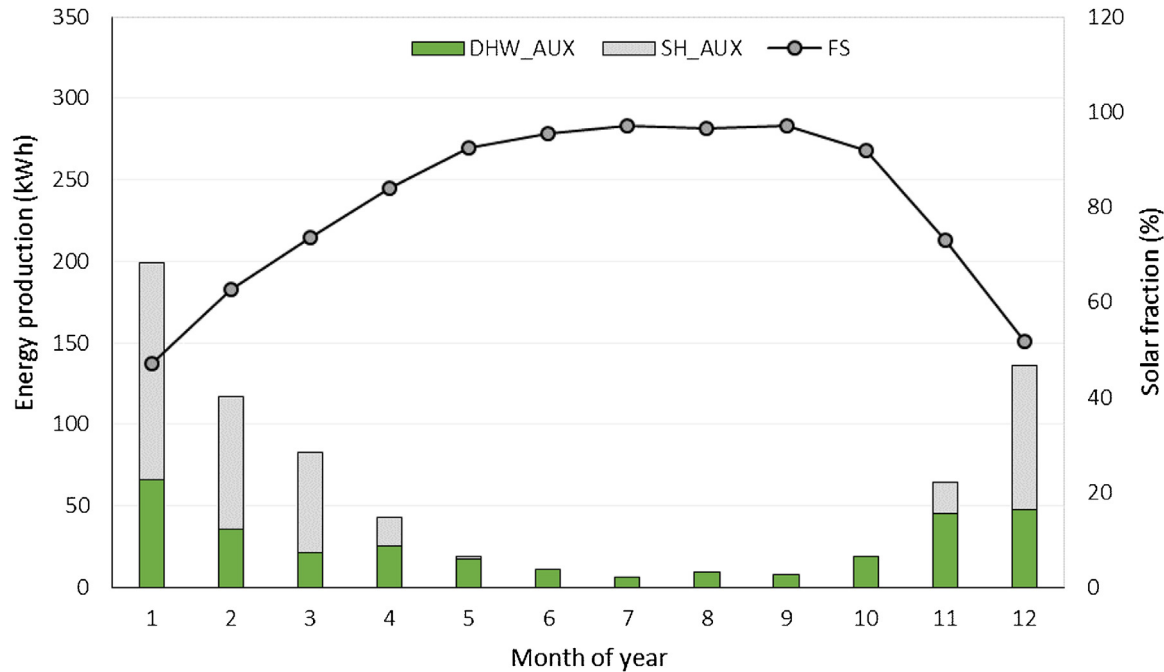


Fig. 15. Monthly heat production of the solar heating system.

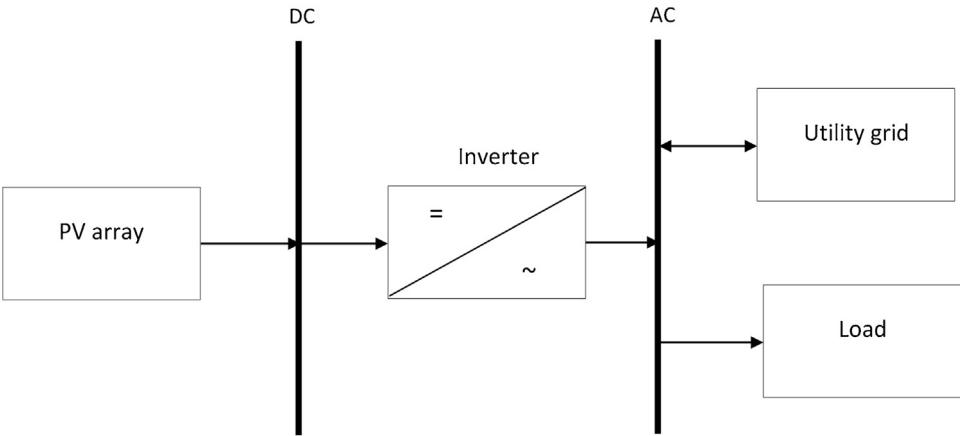


Fig. 16. Grid-connected PV system under study.

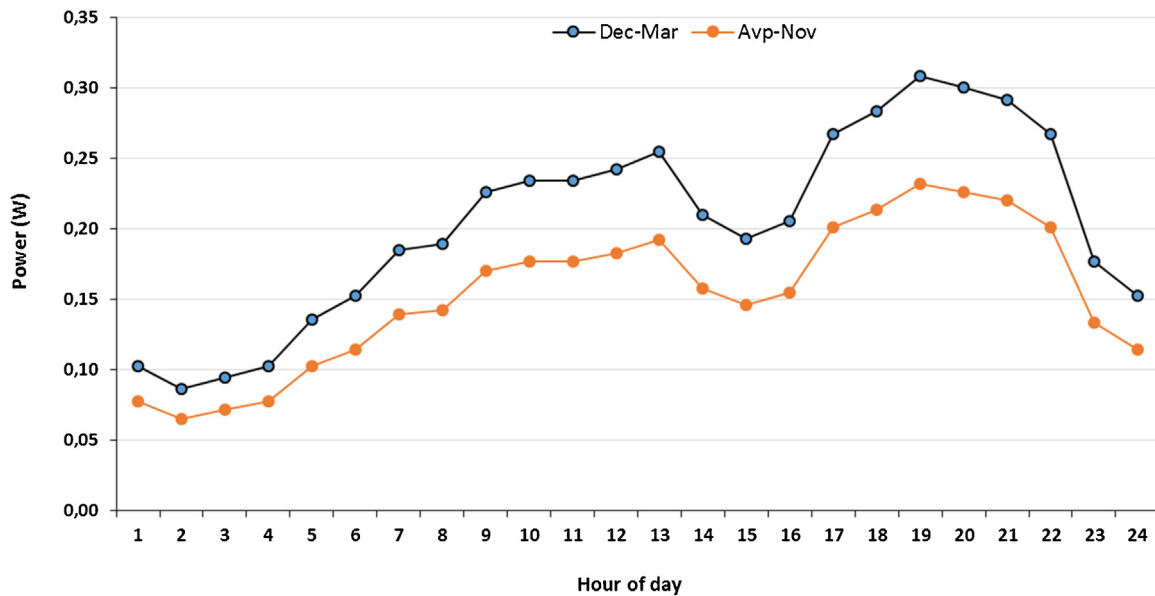


Fig. 17. Daily home electricity consumption profile.

Table 5

The technical specifications of the inverter (SMA 1.3 kW Sunny Boy 1300TL).

Parameter	Specifications
DC Input	
Absolute max voltage	600 V
Absolute max current	11 A
MMPT voltage range	125 V–480 V
AC Output	
Nominal voltage range	180–260 V
Max current	7.2 A
Max power	1300 VA
Efficiency	96%

Table 6

Cost of the solar systems components.

	Number	Unitary cost (USD)	Total (USD)
PV Module (200 W)	4	260	1040
Inverter (800 W)	1	960	960
Electrical energy meter	1	100	100
Thermal collector (2 m ²)	5	640	2300
Storage tank (500 m ³)	1	300	300
Total (USD)			4700

5.1. Electrical consumption profile

The average annual electrical consumption of a single-family house using an efficient electrical equipment is estimated at 1350 kWh [28]. From the annual domestic electrical consumption, the average daily electric load is 3.69 kWh, which is valid from April to November. However, during the heating season (from December to March), in addition to the 1350 kWh for domestic electrical load, the 191 kWh required for SH and DHW are taken into account. Therefore, the annual electrical consumption is 1541 kWh and the average daily electrical load during these months is 5.25 kWh.

Based on these two daily electricity consumption, we perform two hourly domestic electrical load profiles (hour by hour) for these two different periods, which are shown in Fig. 17. We can noticed that electricity consumption is significantly higher in winter than summer due to heating requirements and additional lighting.

5.2. PV system output

The simulation of a grid-connected PV system was undertaken using HOMER, version 2.68 beta [29]. The power produced by the PV system is calculated using the following formula:

$$P_{pv} = P_{max} F_{pv} \left(\frac{G}{G_{STC}} \right) (1 + \alpha (T_c - T_{c,STC})) \quad (3)$$

Where P_{max} is the peak power output (kW), F_{pv} is the derating factor, G is the solar radiation incident on the PV modules in the current hour (kW/m²), G_{STC} is the incident radiation at standard test conditions (1 kW/m²), α is the temperature coefficient of power (%/°C), T_c is the PV cell temperature in the current hour (°C), and $T_{c,STC}$ is the PV cell temperature under standard test conditions (25 °C).

The annual energy balance is obtained by summing the energy interchanges between a grid-connected PV system and utility grid at each hour. The energy balance at any hour of year is given by:

$$E_{PV} + E_{GP} - E_C - E_{GS} = 0 \quad (4)$$

Where E_{PV} is electricity produced by the PV system, E_{GP} is electricity purchased from the utility grid, E_C is energy consumption of the house and E_{GS} is electricity sold to the electricity grid.

The electricity output from the PV system for a specific hour is compared with the corresponding hourly electrical load. If the hourly electrical output exceeds the hourly load, the power is used to satisfy the electrical demand in this hour, and the surplus power is exported to the grid. Thus, the electrical energy purchased from the utility grid in this hour is zero ($E_{GP} = 0$) and the electricity exported to utility grid is by:

$$E_{GS} = E_{PV} - E_C \quad (5)$$

If the hourly electricity output is less than the hourly electrical load, the grid is used to supply the additional power to satisfy the electrical demand in the hour, and no electrical energy is exported ($E_{GS} = 0$). The electricity imported from the grid in this hour is given by:

$$E_{GP} = E_C - E_{PV} \quad (6)$$

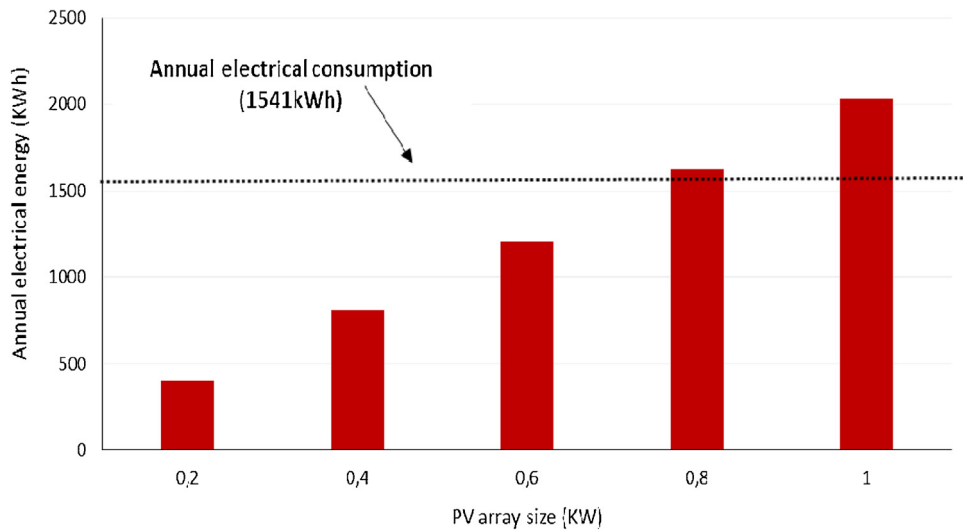


Fig. 18. Annual electricity production with different PV array sizes.

5.3. Optimization of the PV system size

In order to reach the zero energy house, the energy production should be more than the energy demand of the house.

$$EP - EC \geq 0 \quad (7)$$

The energy produced include the thermal energy supplied by the solar heating system and the electrical energy produced by PV system in parallel with the electricity grid.

The energy demand include the heat demand for DHW and space heating and the electricity demand for households and auxiliary electrical heater, as shown in Eq.

$$(EP_{shs} + EP_{pv}) - (EC_{households} + EC_{elec.heater} + EC_{sh} + EC_{dhw}) \geq 0 \quad (8)$$

Heat demand for DHW and space heating is covered by the solar heating system. Thus, for achieved a zero energy house:

$$EP_{pv} \geq EC_{households} + EC_{elec.heater} \quad (9)$$

Therefore, in order to design a PV system according to the electricity load required for the reference house, we have studied the energy performance of different PV array sizes (from 0.4 to 2 kW_p).

As shown in Fig. 18, a PV array of 1.6 kW_p may produce more than the annual electrical consumption of the reference house

(2539 kWh). As a result, a 1.6 kW_p PV system is selected for analysis in this study. The designed PV array is composed of nine PV modules, each panel produces 36.5 V and 200 W_p (5.48 A).

5.4. Simulation results of 0.8 kW_p PV system

The annual electrical simulations results for a 0.8 kW_p PV system are given in Fig. 19. The total electricity consumption in the home from all sources “AC primary load”, supplied in parallel from the inverter and the network is 1622 kWh and the total annual kWh exported to the electricity network per annum which occurs when the PV inverter generates excess electricity relative to the home electricity demand “grid sales” is 920 kWh. The total annual kWh imported from the electricity network per annum which occurs when the PV inverter generates less electricity relative to the home electricity demand “Grid purchases” is 740 kWh and the total annual kWh produced by the PV system “PV array” is 1622 kWh. The PV system produced the equivalent of 70% of the total electricity consumed in the home. The annual electricity purchased from the grid represents 30% of the total electricity consumption in the home.

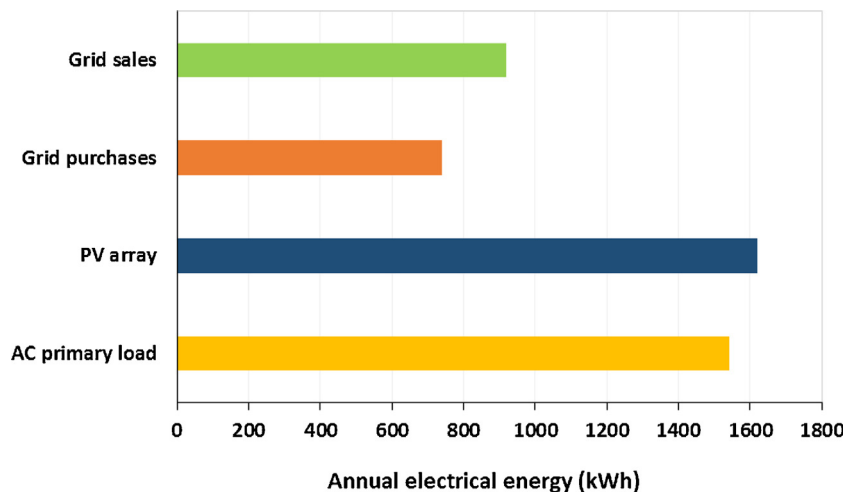


Fig. 19. Electrical simulation results.

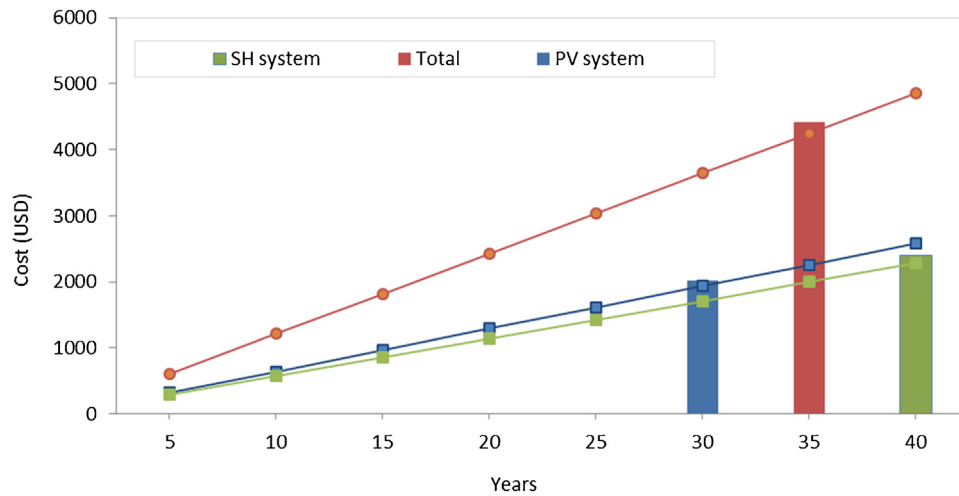


Fig. 20. Capital cost of solar systems and their investment return.

5.5. Economic analysis

In order to carry out economic analysis due to the integration of a grid connected PV system and solar heating system in bioclimatic house, the estimation of the yearly income and the cost of each component of the systems must be set out. In the analysis, the time value of money, the effective interest rate or rising energy prices, or the replacement cost where not considered. The life cycle of the solar systems components is estimated to be 30 years. To compile the most accurate and realistic prices for the case study house, every effort was made to get up-to-date pricing from local vendor for the solar components.

5.5.1. Yearly income

The yearly income due to electricity and gas saving for the studied house when the grid connected PV system and the solar heating system are installed was calculated as follows:

$$YI = YI_{elec} + YI_{gas} \quad (10)$$

The costs savings due to the thermal and electrical energy savings are calculated according to the following Eqs. (11) and (12):

$$YI_{gas} = SF \times ED_{heat} \times C_g \quad (11)$$

Here, ED_{heat} is the yearly thermal energy demand and C_{gas} , stands for the price of electricity purchased from the electricity grid assumed to be 1.1 DZA/kWh (0.01 USD/kWh).

$$YI_{elec} = ED_{el} \times C_{el} + (ED_{heat} - SF \times ED_{heat}) \times C_{el} \quad (12)$$

Where, ED_{el} is the yearly electricity demand and C_{el} , stands for the price of electricity purchased from the electricity grid assumed to be 4.179 DZA/kWh (0.04 USD/kWh).

Note that the yearly electricity demand includes the auxiliary energy from direct electrical heating in addition to the domestic electrical demand (lighting and appliances).

5.5.2. Initial cost of PV system

Market cost of the grid PV system IC_{PV} includes the cost of the PV array and the inverter, as shown in Table 6. The costs for mounting the PV system is assumed to be 10% of the investment cost for the PV system. Hence,

$$IC_{PV} = 1.1 \times (N_{module} \times C_{array} + C_{inverter} + C_{meter}) \quad (13)$$

Where, N_{module} represents the number of module of array, C_{array} represents the average price of the module per the unit of its rated

power and $C_{inverter}$ represents the average price for the inverter, and C_{meter} represents the cost of the electrical energy meter.

The cost of the solar heating system IC_{sh} include the cost of thermal collectors and storage tank.

$$IC_{sh} = 1.2 \times (N_{module} \times C_{collector} + C_{tank} + C_{floor}) \quad (14)$$

Where, $C_{collector}$ is the average price of the thermal collector, C_{tank} the price of the storage tank and C_{floor} the price of the heating floor system. The installation and the floor heating system costs are assumed to be 20% of the investment cost for the solar heating system.

5.5.3. Return on investment

The return on investment is defined as the initial investment cost divided by the annual cost savings due to the installation of solar system.

$$RI = \frac{IC}{YI} \quad (15)$$

Fig. 20 represents the investment cost in the grid connected PV system and solar heating system and their corresponding return on investment. The total investment in solar systems require a high initial cost (4700 USD) leading to high return on investment (35 years). This is due mainly to the high cost of solar energy components and the low price of the electricity generated from the utility grid.

The investment in solar heating system is high than the investment in grid connected PV system due the high thermal collector's areas required for space heating and DHW preparation. Therefore, the return on investment is, 30 and 40 years for, respectively, the PV system and the solar heating system.

From Fig. 21, it can be noticed that the solar systems become economically feasible if one of the following solutions are adopted by the Algerian government:

- Multiply the price of gas by 3 and electricity by 2.
- Subsidises 50% of the installations cost.

In the case where the price of the conventional energy is increased, the investment is returned after less than 25 years. While, if the authorities subdivides 50% of the installation cost, the return on investment decreases from 35 to about 20 years.

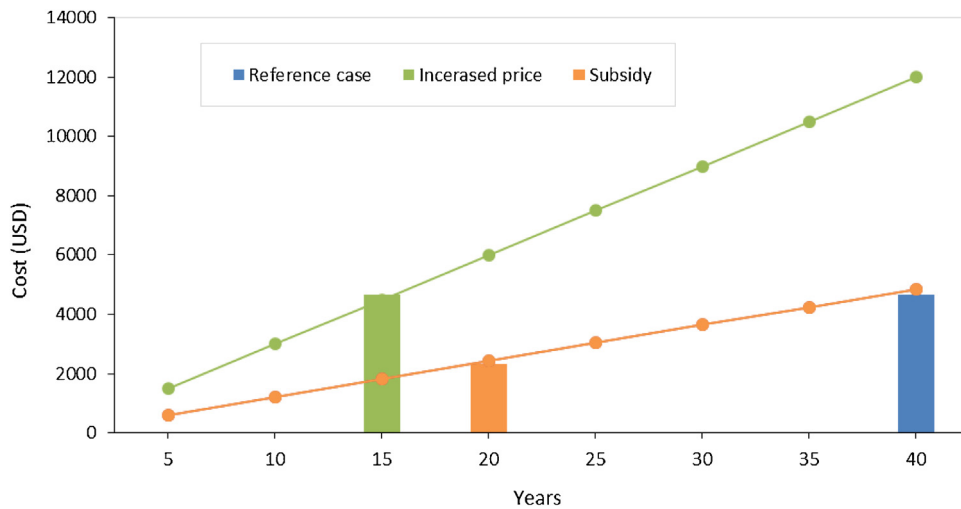


Fig. 21. Return on investment for different cases.

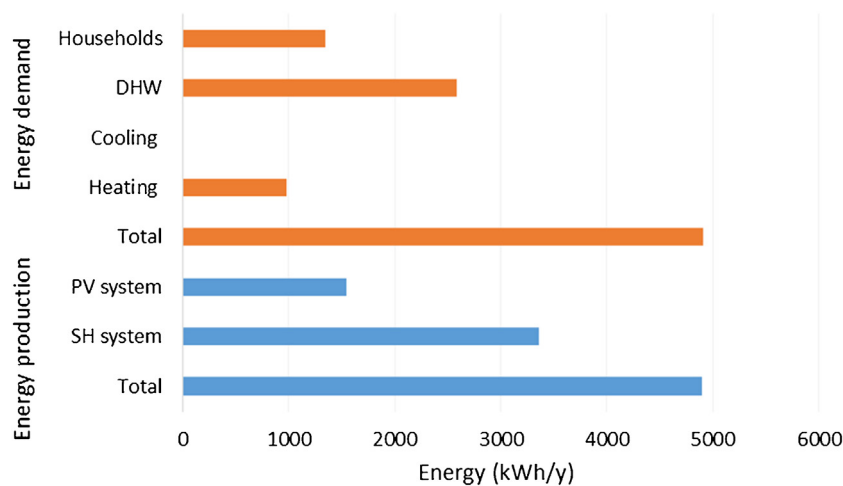


Fig. 22. Comparison between annual energy demand and energy production.

Table 7
Comparison between energy demand and energy production.

Energy demand		Energy production	
Households	1350	Solar heating system	2374
Space heating	982	PV system	2599
Space cooling	0		
DHW preparation	2581		
Total	4913	Total	4973

6. Conclusion

The aim of this work was to investigate the feasibility of zero energy house in the north of Algeria by application of a grid connected PV system in a bioclimatic house prototype. From the experimental results, it is shown that bioclimatic designs integration, such that adequate orientation of the opening, well insulation of the house envelope and using solar protection and application of night ventilation in summer, led to a significant improvement in the indoor comfort. Therefore, the simulation results show that the energy demand of the bioclimatic house is very low, which allow the achievement of the goal of zero energy status by integration of a grid connected PV system, in addition to the solar heating system, very easily. The annual energy demand and generation is summarized in Fig. 22 and Table 7. The total energy demand for domestic

electricity, DHW, and space heating and cooling can be completely produced by solar heating system and PV system in parallel with the utility grid. The energy required for space heating and DHW preparation are 982 kWh and 2581 kWh, respectively. A solar heating system with 10 m² of flat-plate collector area and 500-l storage tank provide about 93% of total heat demand. The rest of the heat demand (7%) and domestic electricity can be provided by an auxiliary electrical heater, which is supplied by 0.8 kW_p PV system in parallel with the utility grid.

From this work, the following recommendations could be made:

- Minimizing the energy requirements through efficient design is the first criteria to achieve the goal of a zero energy house. The remain needs will be covered by renewable energy generation.
- The size of the solar heating system components should be optimized in order to meet the maximum of thermal demand of the house.
- The use of a PV system in parallel with the electricity grid represents the best solution to reach zero energy house status.
- Due to the low current prices of conventional energy, the investment in solar energy systems is not economically profitable. However, the system could be profitable if the government subsidises the capital cost and rises conventional energy prices.

References

- [1] MEM, Ministère de l'énergie et des mines. Agence Nationale pour la Promotion et Rationalisation de l'Utilisation de l'énergie (APRUE). Consommation Energétique Finale de l'Algérie, 2009. WWW.aprue.org.dz.
- [2] CNERIB, 'Document Technique Règlementaire, "Règlement Thermique des Bâtiments d'Habitation – Règles de Calcul des Déperditions Calorifiques", DTR C3-2', Fascicule 1, in: Rapport Technique, CNERIB, Alger, 1998.
- [3] CNERIB, 'Document Technique Règlementaire, "Règlement Thermique des Bâtiments d'Habitation – Règles de Calcul des Apports Calorifiques des Bâtiments", DTR C3-4', Fascicule 2, in: Rapport Technique, CNERIB, Alger, 1998.
- [4] CNERIB, 'Document Technique Règlementaire, "Règlement Thermique des Bâtiments d'Habitation – Ventilation naturelle des locaux à usage d'habitation", DTR.C. 3-31', fascicule 3, in: Rapport Technique, CNERIB, Alger, 1998.
- [5] MEM. (Ministère de l'Energie et des Mines), Programme des Energies Renouvelables et de l'Efficacité Energétique, Algérie, 2011.
- [6] Jordi Llovera, Xavi Potau, Marc Medrano, Luisa F. Cabeza, Design and performance of energy- efficient solar residential house in Andorra, *Appl. Energy* 88 (2011) 1343–1353.
- [7] Samar Jaber, Salman Ajib, Optimum technical and energy efficiency design of residential building in Mediterranean region, *Energy Build.* 43 (2011) 1829–1834.
- [8] Danielle Griego, Moncef Krarti, Abel Hernández-Guerrero, Optimization of energy efficiency and thermal comfort measures for residential buildings in Salamanca, Mexico, *Energy Build.* 54 (2012) 540–549.
- [9] Naouel Daouas, A study on optimum insulation thickness in walls and energy savings in Tunisian buildings based on analytical calculation of cooling and heating transmission loads, *Appl. Energy* 88 (2011) 156–164.
- [10] I. Pyeongchan, K. Moncef, Design optimization of energy efficient residential buildings in Tunisia, *Build. Environ.* 58 (2012) 81–90.
- [11] B. Viorel, Case study for active solar space heating and domestic hot water preparation in a passive house, *J. Renew. Sustain. Energy* 3 (2011) 023102.
- [12] Guilherme Carrilho da Graça, Andrei Augusto, Maria M. Lerer, Solar powered net zero energy houses for southern Europe: feasibility study, *Sol. Energy* 86 (2012) 634–646.
- [13] B. Milorad, N. Novak, N. Danijela, S. Jasmina, M. Ivan, Toward a positive-net-energy residential building in Serbian conditions, *Appl. Energy* 88 (2011) 2407–2419.
- [14] W. Liping, G. Julie, J. Phil, Case study of zero energy house design in UK, *Energy Build.* 41 (2009) 1215–1222.
- [15] S.M. Bambrook, A.B. Sproul, D. Jacob, Design optimisation for a low energy home in Sydney, *Energy Build.* 43 (2011) 1702–1711.
- [16] L. Mitchell, Z. Radu, Life cycle cost and energy analysis of a Net Zero Energy House with solar combisystem, *Appl. Energy* 88 (2011) 232–241.
- [17] TRNSYS, User's Manual, A Transient System Simulation Program Version 16, Solar Energy Laboratory, University of Wisconsin-Madison, WI, USA, 2006.
- [18] HOMER Energy LLC, 2010. HOMER Version, 2.68 beta. National Renewable Energy Laboratory, Operated for the U.S. Department of Energy Office of Energy Efficiency and Renewable Energy, 2010.
- [19] Alexandre Hugo, et al., Solar combisystem with seasonal thermal storage, *J. Build. Perform. Simul.* 3 (December (4)) (2010) 255–268.
- [20] Shilei Lua, et al., Establishment and experimental verification of PCM room's TRNSYS heat transfer model based on latent heat utilization ratio, *Energy Build.* 84 (2014) 287–298.
- [21] F. Sahnoune, M. Belhamel, M. Zemat, Etude comparative des performances thermiques d'un prototype de chauffe-eau solaire pour deux sites algériens, *Revue des énergies renouvelables* 14 (3) (2011) 481–486.
- [22] K. Imessad, et al., Impact of passive cooling techniques on energy demand for residential buildings in a Mediterranean climate, *Renew. Energy* 71 (2014), 589e597.
- [23] Alireza Hobbi, Kamran Siddiqui, Optimal design of a forced circulation solar water heating system for a residential unit in cold climate using TRNSYS, *Sol. Energy* 83 (2009) 700–714.
- [24] W.E. Buckles, S.A. Klein, Analysis of solar domestic hot water heaters, *Sol. Energy* 25 (5) (1980) 417–424.
- [25] S.A. Klein, W.A. Beckman, J.A. Duffie, A design procedure for solar heating systems, *Sol. Energy* 18 (1976) 113.
- [26] W.A. Beckman, S.A. Klein, J.A. Duffie, *Solar Heating Design By the F-Chart Method*, John Wiley, New York, 1977.
- [27] J.W. Baughn, M.F. Young, The calculated performance of a solar hot water system for a range of collector flow rates, *Sol. Energy* 32 (2) (1984) 303–305.
- [28] M. Missoum, et al., Impact of rural housing energy performance improvement on the energy balance in the North-West of Algeria, *Energy Build.* 85 (2014) 374–388.
- [29] Zhe Li, F. Boyle, A. Reynolds, Domestic application of solar PV systems in Ireland: the reality of the their economic viability, *Energy* 36 (2011) 5865–5876.